**Submitted by : Maha Munir Mohamed "100523733077" ,**
**Hrudya Mashetty "100523733064"**
**Sapavat Lokeshwari "100523733052"**

# Problem Formulation: Grid Robot Navigation

Project 14 | Reinforcement Learning Project.

Objective: Train a simulated robot to reach a goal in a grid maze while avoiding obstacles and minimizing unnecessary moves.

Environment: A deterministic 4 × 4 grid. The start cell is (0, 0), the goal is (3, 3), and blocked cells are (1, 1) and (2, 1). Attempting to move beyond the grid or into a blocked cell leaves the agent in place. An episode ends when the goal is reached or after 50 actions.

State: The agent's current grid location, represented by its row and column (row, column). Because the maze and its rules are fixed, this state fully describes the agent's position.

Actions: Move right, left, down, or up. Invalid moves receive a penalty and do not change the agent's location.

Reward: +10 for reaching the goal, −1 for each ordinary move, and −5 for an invalid move. The goal reward encourages completion, the step cost favors shorter routes, and the invalid-move penalty discourages collisions with walls and obstacles.

Algorithm: Tabular Q-learning with learning rate $\alpha = 0.20$ and discount factor $\gamma = 0.95$. Train for 1,000 episodes with ε-greedy action selection ($\varepsilon = 0.20$). The update is $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max Q(s',a') - Q(s,a)]$. The baseline chooses actions randomly.

Evaluation: Record mean return in consecutive 50-episode windows and compare Q-learning with a random policy on the same maze and step limit. Evaluate the final greedy Q-policy for 200 episodes. Results apply to this small fixed simulation; they do not establish performance in a changing or physical environment.